

## LIGHTWEIGHT YOLOX-BASED GREEN ONION BRANCHING POINT DETECTION FOR AUTOMATED PEELING ON EDGE DEVICE

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**ABSTRACT.** *Agricultural automation has become critical as labor continues to decline worldwide. In green onion processing, current peeling machines require significant manual work due to imprecise leaf removal. Prior edge detection methods were tested only on uniform stems of a single variety, limiting their practical use where green onions vary in morphology. The key challenge is developing a detection system that maintains performance across varieties and growth stages without frequent adjustments. Here we show that an improved YOLOX-nano model effectively detects branching points across diverse varieties, achieving a 0.929 detection score for the second branching point and 46 ms processing time on Jetson Orin Nano. Unlike edge detection methods, our deep learning approach handles variations in onion varieties and growth stages robustly and maintains performance under varying conditions. These results show that lightweight deep learning systems can reliably automate green onion peeling while achieving real-time performance on edge devices. The proposed system advances automated green onion peeling and could extend to similar agricultural tasks.*

**Keywords:** Object detection, YOLOX, Edge device, Smart agriculture, Green onion

**1. Introduction.** In recent years, the labor shortage in agricultural production fields has become severe in many developed countries [1, 2]. Therefore, automating agricultural production using IT technology has become an urgent task. In agricultural production fields in Japan, in addition to the continuous decrease of farm workers, the rapid aging of workers has become a particularly serious problem. In response to these problems, the demand is increasing for automating agricultural machinery to reduce labor, and various developments are underway [3, 4, 5].

In green onion peeling facilities in Japan, labor shortage is also critical. When preparing green onions for shipping, they must retain “one stem and one leaf” to maintain freshness. However, current peeling machines lack precision and cannot remove all unwanted leaves in one process. As a result, workers must manually remove the remaining leaves after machine processing.

Given the critical labor shortage, developing automated peeling machines to reduce manual work has become imperative. Our analysis reveals that precise peeling requires careful alignment of air pressure nozzles with the green onion’s second branching point.

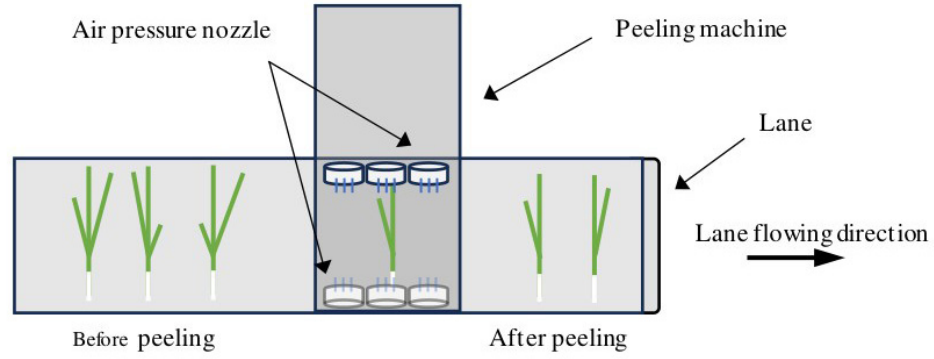


FIGURE 1. The overview of the green onion peeling machine

As shown in Figure 1, green onions enter the machine via a conveyor belt where nozzles perform the peeling process. However, conventional systems employ fixed nozzle positions that cannot adapt to natural variations in branching points between individual plants. To enable accurate automated peeling, we require an intelligent system capable of detecting the second branching point and dynamically adjusting nozzle positions. For practical implementation, the system must achieve a 90% success rate while completing each peeling operation within 50 ms.

In our previous work, we proposed an edge detection based methodology for green onion branching point detection [6]. The algorithm identifies branching points by edge detection to detect the characteristic diagonal lines at branching points on the leaf. Experimental evaluation on a Raspberry Pi 3 demonstrated a detection rate of 90.6% for branching point diagonal lines with a processing latency of 455 ms. However, this method faces two significant challenges. First, detection consistency decreases because green onions' physical characteristics vary between cultivars and stem size, making uniform detection difficult. Second, leaf obstructions frequently cause detection errors, requiring parameter tuning periodically to maintain reliable performance.

In recent years, image recognition work for agricultural products has progressed significantly, and researchers have proposed numerous DNN based methods [7, 8, 9]. Among DNNs, YOLO achieves high-speed object detection by simultaneously detecting object locations and classes in a single network. This capability has made YOLO widely adopted in agricultural machinery and robotics applications on edge devices [10, 11, 12]. Building on these advances, we propose a novel two-stage detection system using a lightweight YOLOX-nano model optimized for edge deployment.

The objective of this work is to develop a robust and efficient system for green onion peeling. This system must maintain high accuracy across green onions of different varieties and growth stages. Additionally, it needs to achieve real-time performance on edge devices. The contributions of this paper can be summarized as follows. First, we develop a two-stage system that handles variety differences by classifying branching points into three classes (First, Other, and Aging) and using a robust selection algorithm. Second, we propose a lightweight YOLOX-nano model by strategic layer reduction for efficient processing while maintaining accuracy across varieties, stem sizes, and growth stages. Third, we demonstrate viability by implementing the system on Jetson Orin Nano, achieving 0.929 detection score and 46 ms processing time in 7.5 W mode, meeting real-world requirements.

We organize the remainder of the paper as follows. In Section 2, we present our lightweight YOLOX-nano model and its optimization for green onion branching point detection. Section 3 shows the experimental results. Finally, we conclude in Section 4 with a summary and mention of future work.

**2. Proposed Method.** In this section, we describe the overview of the proposed method and the lightweight model design. We propose a system that detects the second branching point of green onions from images. As shown in Figure 2, the second branching point of a green onion is the position of the second branch counted from the tip of the leaf. We aim to automatically adjust the green onion peeling position by detecting the second branching point using the YOLOX model to extract its  $y$ -coordinate position.

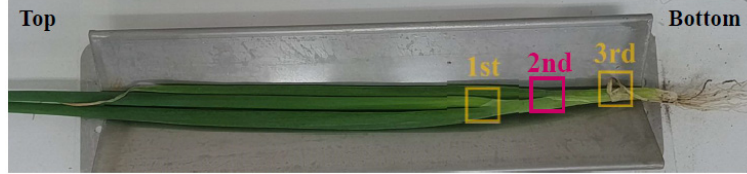


FIGURE 2. Example of green onion branching points

**2.1. Second branching point identification algorithm.** The system consists of two steps: the detection step and the identification step. First, in the detection step, the YOLOX model detects the branching points of green onions. The YOLOX model detects three classes: First (first branching point), Other (branching points after the second), and Aging (dead leaf branching points). Next, in the identification step, the system identifies the second branching point from the detected branching points. Specifically, if objects of the Other class are detected, it selects the highest one among them. If no Other class objects are detected, it selects the highest one among the Aging class. Finally, the  $y$ -coordinate position of the identified second branching point is output.

**2.2. YOLOX model.** Next, we introduce the YOLOX model used in this system. YOLOX is an object detection algorithm proposed and is based on YOLOv3 using Darknet53 with many design improvements [13]. The model achieves exceptional performance in both speed and accuracy, making it particularly suitable for industrial applications. Its architecture consists of three main components. First, the backbone network uses CSPNet (Cross Stage Partial Network) [14] for feature extraction and includes the SPP-Bottleneck module [15] for effective multiscale learning. Second, the neck utilizes PAFPN (Path Aggregation Feature Pyramid Network) [16] to integrate features across multiple scales. Finally, the model processes input images through these networks and splits them into three scales, using a Decoupled head design that separately predicts object class and location to enhance detection accuracy.

**2.3. Lightweight method.** Figure 3 shows the architecture of YOLOX with the lightweight methods applied. To optimize the model for our specific use case, we implemented three lightweight methods (RLL, RML, and RBL). The original YOLOX architecture utilizes a Feature Pyramid Network (FPN) structure that outputs predictions at three different scales: stride 8 (high-resolution), stride 16 (middle-resolution), and stride 32 (low-resolution), to detect small, medium, and large objects, respectively. To optimize the model for our specific use case, we implemented three lightweight methods: Reduction of Low-resolution Layer (RLL), Reduction of Middle-resolution Layer (RML), and Reduction of Bottleneck Layer (RBL). First, regarding RLL and RML, we focused on the physical characteristics of the target. Since green onion branching points appear as relatively small, localized features within the image, the high-level semantic information from deeper layers is less critical than the fine-grained spatial details found in the shallow layers. Specifically, RLL removes the prediction head and feature maps for stride 32 (for large objects), and RML removes those for stride 16 (for medium objects). By retaining only the high-resolution layer (stride 8), we significantly reduced the computational cost and memory usage without sacrificing localization accuracy for small branching points.

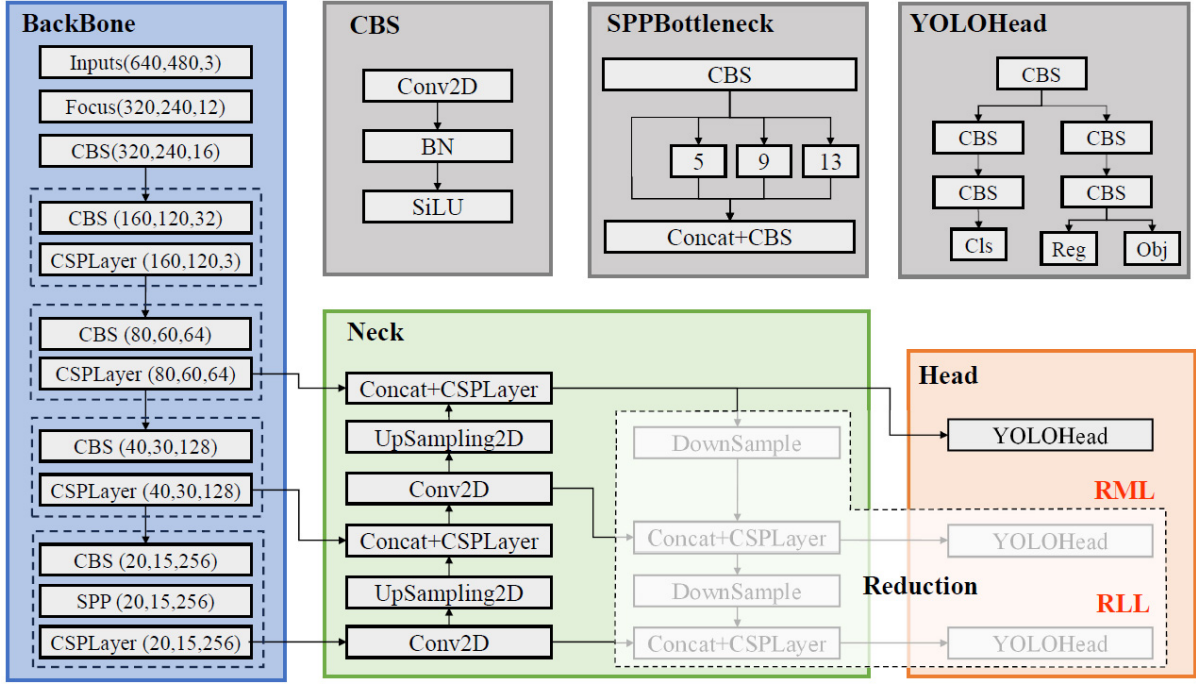


FIGURE 3. YOLOX architecture with lightweight methods

For the RBL (Reduction of Bottleneck Layer), we simplified the CSP modules in the backbone. Since the visual features of green onion stems are geometrically simple compared to general objects (e.g., animals or vehicles), excessive depth in feature extraction is redundant. By reducing the number of bottleneck blocks, we successfully balanced high-speed inference with sufficient feature representation capability.

**3. Experiments and Results.** This section introduces the overview of the experiments and results. The purpose of this experiment is to evaluate the effectiveness of the proposed system using YOLOX for the detection of the second branching point in terms of detection accuracy and processing time.

**3.1. Experimental design and methodology.** We conducted the training on a PC with Ubuntu 20.04, equipped with an Intel Core i5-11400 CPU and NVIDIA GeForce RTX 3060 GPU. The implementation used Python 3.8, PyTorch 1.9.0, and CUDA 11.1. The model was trained for 300 epochs with a batch size of 16, using the Adam optimizer with a learning rate of 0.001.

The green onion dataset consists of images captured at the peeling facility in Usa City, Oita Prefecture, Japan. As shown in Figure 4, it contains 806 images with a resolution of  $1980 \times 1080$  px, featuring three varieties: Power Slim, Wakatsuki, and Dark Slim. While previous work used datasets limited to M-size (stem diameter 5.1-6.0 mm) Power Slim variety, our dataset includes green onions across all size categories: S (up to 5.0 mm), M (5.1-6.0 mm), and L (over 6.1 mm) as defined by JA (Japan Agricultural Cooperatives) shipping standards. This variety better represents real-world processing scenarios. We utilized 565 images for training and 241 images for testing, with labels annotated into three classes: First, Other, and Aging. To address data imbalance, we performed K-fold cross-validation, dividing the dataset into K subsets, using one as test data and the others for training. This process was repeated K times, allowing all data to be evaluated. In this experiment, we adopted 5-fold cross-validation ( $K = 5$ ) and averaged the results across all folds.

The system evaluation was performed on an NVIDIA Jetson Orin Nano. This edge device features an Ampere GPU and 8-core ARM CPU, delivering 40 TOPS compute

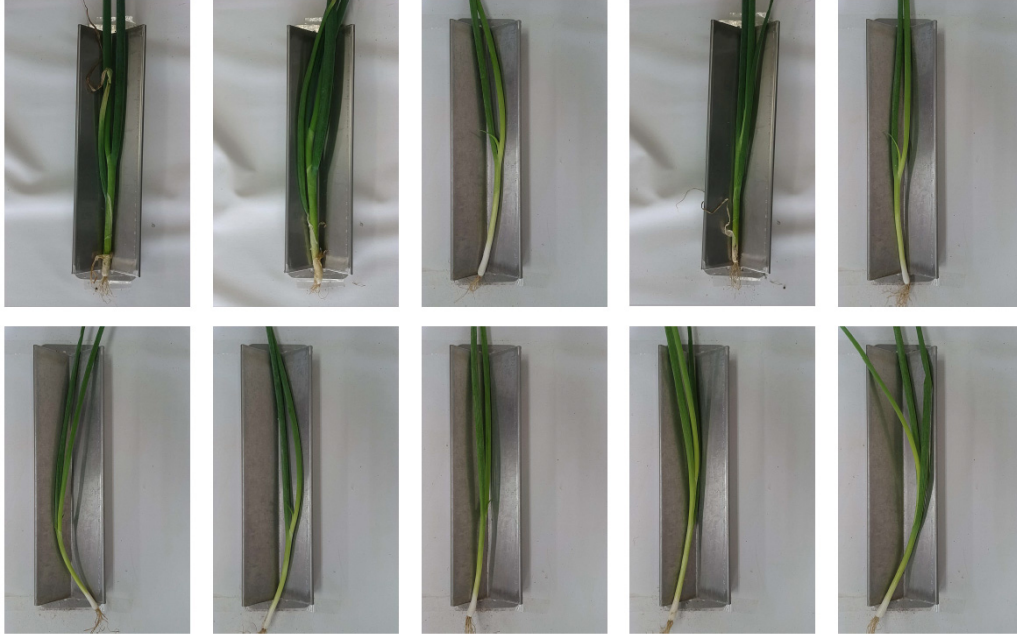


FIGURE 4. Sample images from the green onion dataset

power. We implemented the system in 7.5 W mode instead of 15 W to save power for real-world deployment.

**3.2. Evaluation metrics.** Our system has two steps: the detection step and the identification step. We measure detection accuracy using Precision, Recall, AP, and mAP metrics. In the context of branching point detection, the terms used in these metrics are defined as follows. True Positive (TP) refers to a correctly detected branching point. False Positive (FP) refers to a non-branching point (e.g., a leaf edge or overlap) incorrectly detected as a target. False Negative (FN) refers to an actual branching point that the model failed to detect. Based on these definitions, Precision and Recall are calculated as

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (1)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (2)$$

Furthermore, we use Average Precision (AP) to evaluate the overall performance.

$$\text{AP} = \int_0^1 p(r) dr \quad (3)$$

where  $p(r)$  represents the precision value at a specific recall level  $r$ . This integral calculates the area under the Precision-Recall (PR) curve, summarizing the model's performance across all confidence thresholds. Additionally, we use mAP (mean Average Precision) to evaluate the overall performance.

$$\text{mAP} = \frac{1}{n} \sum_{i=1}^n \text{AP}_i \quad (4)$$

In our previous work, we evaluated branching point detection accuracy simply by checking whether the positional error between ground truth and predicted positions was within 1 cm. However, positioning the nozzle above the true branching point increases the risk of damaging upper leaves and causing peeling failures. Moreover, the peeling success rate decreases as the deviation between predicted and actual branching positions increases.

To address these factors, we introduced a Distance Score metric that evaluates the positional error between predicted and actual branching points. The Distance Score is defined based on the error distance ( $\Delta y_{cm}$ ) between the ground truth position ( $y_t$ ) and predicted position ( $y_p$ ):

$$\text{score} = \begin{cases} 1 & (y_p < y_t, \Delta y_{cm} < 0.5) \text{ or } (y_p \geq y_t, \Delta y_{cm} < 1.0) \\ \max\left(0, 1 - \frac{\Delta y_{cm} - 0.5}{0.5}\right) & (y_p < y_t, 0.5 \leq \Delta y_{cm}) \\ \max\left(0, 1 - \frac{\Delta y_{cm} - 1}{2.0}\right) & (y_p \geq y_t, 1.0 \leq \Delta y_{cm}) \end{cases} \quad (5)$$

This scoring method is designed based on experimental observations of peeling operations using high-pressure nozzles. When the predicted position is above the correct position ( $y_p < y_t$ ), an error of less than 0.5 cm is acceptable (score = 1), and in the range from 0.5 cm to 1.0 cm, a linear penalty is applied. On the other hand, when the predicted position is below the correct position ( $y_p \geq y_t$ ), an error of less than 1.0 cm is acceptable (score = 1), and in the range from 1.0 cm to 3.0 cm, a linear penalty is applied. Moreover, errors beyond these ranges lead to failure in peeling operations, resulting in the maximum penalty (score = 0).

**3.3. Analysis of experimental results.** We compared different YOLOX model sizes to evaluate our lightweight YOLOX-nano model. We evaluated the inference accuracy and inference time in the detection step including three classes: First, Other, and Aging. Additionally, we compared our lightweight YOLOX-nano model with different YOLOX model sizes. Table 1 shows the evaluation results in terms of precision, recall, mAP@0.5, and processing time. Our proposed YOLOX-nano model achieved competitive precision and recall scores of 0.826 and 0.865, respectively, along with an mAP of 0.494, demonstrating its effectiveness despite being the lightest model. This indicates that complex feature extraction is not necessary for green onion branching point detection. Furthermore, the inference processing time was 29.37 ms, making it the fastest among all models tested.

TABLE 1. Performance comparison of YOLOX models

| Model                        | Evaluation metrics |              |              | Processing time [ms] |
|------------------------------|--------------------|--------------|--------------|----------------------|
|                              | Recall             | Precision    | mAP@0.5      |                      |
| <b>YOLOX-nano (proposed)</b> | <b>0.865</b>       | <b>0.826</b> | <b>0.494</b> | <b>29.37</b>         |
| YOLOX-Tiny                   | 0.896              | 0.861        | 0.496        | 38.05                |
| YOLOX-S                      | 0.904              | 0.865        | 0.493        | 46.35                |
| YOLOX-M                      | 0.898              | 0.865        | 0.499        | 104.35               |

Regarding the selection of the base architecture, although newer models like YOLOv8 exist, we prioritized parameter efficiency over marginal accuracy gains. YOLOX-nano (0.91 M parameters) is significantly lighter than YOLOv8-nano (approx. 3.2 M parameters). Applying such heavier models to geometrically simple targets like green onions would be over-engineering. Our results in Table 1 confirm that the optimized YOLOX model provides the “right-sized” solution, fulfilling the strict latency requirement (< 50 ms) with sufficient accuracy.

Next, we evaluated the identification accuracy of branching points in the identification step and the processing time of the entire system. This is because if the detection accuracy of the first branching point is low, the detection accuracy of the second branching point will also decrease. Figure 5 presents a comparison of Distance Scores and overall processing times. The model achieved a Second branch Distance Score of 0.929, confirming its high

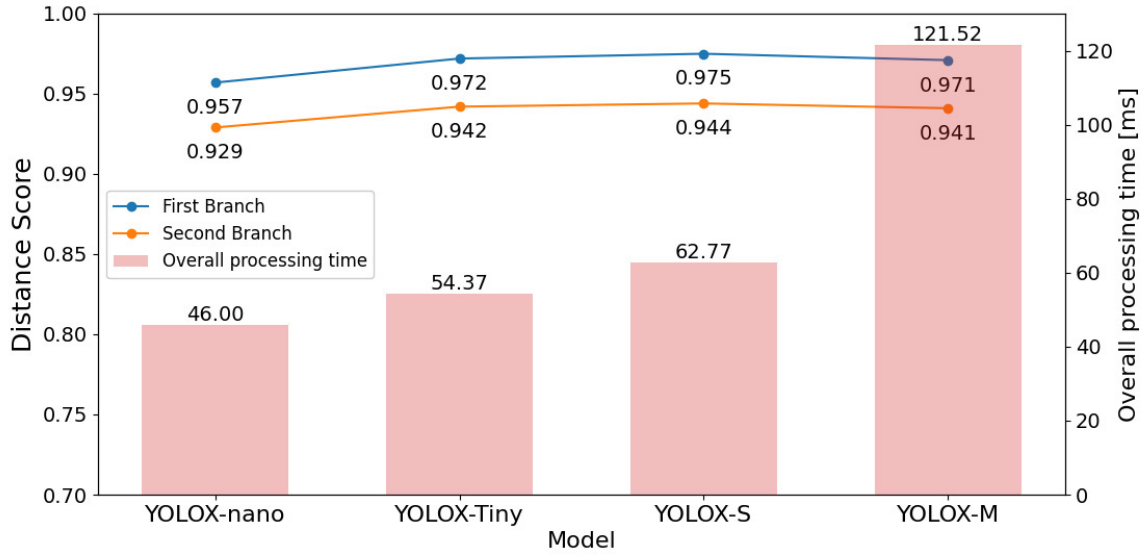


FIGURE 5. Distance Score and processing time comparison

detection accuracy. The total system processing time, which includes image preprocessing and post-processing steps, was 46.00 ms. This is well below our target of 50.00 ms, demonstrating practical processing speeds suitable for production environments. These results validate that our proposed method delivers both robust detection accuracy across varying environmental conditions and the high-speed processing capabilities required for production line implementation.

Figure 6 illustrates qualitative results of the detection. As shown in Figure 6(a), our method accurately detects the second branching point across green onions with varying stem thicknesses and curvatures, demonstrating the model's robustness to morphological variations. However, Figure 6(b) highlights failure cases, particularly in dense leaf clusters. These errors primarily stem from visual ambiguity caused by overlapping leaves. When leaves overlap, they often create V-shaped shadows or intersecting edges that visually resemble the characteristic shape of a branching point. This similarity leads the model to produce False Positives (misidentifying a leaf overlap as a branching point). Additionally, severe occlusion where a foreground leaf completely covers the branching point can result in False Negatives (failing to detect a branching point). This analysis suggests that while the lightweight YOLOX model is effective for general morphology, it struggles with complex occlusion patterns.

**3.4. Ablation study.** We conducted an ablation study to evaluate three simplification methods: Reduction of Low-resolution Layers (RLL), Reduction of Middle-resolution Layers (RML), and Reduction of Bottleneck Layers (RBL). Table 2 indicates that the processing time decreased by 4.1% from 30.63 ms to 29.37 ms compared to the baseline, while maintaining critical accuracy in detecting the second branching point with a Distance Score of 0.918 and 0.929. The overall mAP showed a marginal decrease from 0.496 to 0.494, indicating our modifications preserve essential detection capabilities while improving efficiency.

**4. Conclusions.** In this paper, we proposed a lightweight YOLOX-nano based system for robust branching point detection across diverse green onion varieties (Power Slim, Wakatsuki, Dark Slim), stem sizes, and growth stages. Our architecture optimizations addressed critical challenges of morphological variation and leaf occlusion while maintaining computational efficiency. Experimental validation on the Jetson Orin Nano showed real-time performance with a processing time of 46 ms and a power consumption of 7.5



(a) Successful detection: Accurate detection of the second branching point



(b) Failed detection: Misdetection in dense leaf areas

FIGURE 6. Examples of successful and failed detections using YOLOX

TABLE 2. Ablation study of YOLOX-nano

| Model       | Ablation elements |     |     | Distance Score |              | mAP          | Processing time [ms] |
|-------------|-------------------|-----|-----|----------------|--------------|--------------|----------------------|
|             | RLL               | RML | RBL | First          | Second       |              |                      |
| YOLOXN-base | ✗                 | ✗   | ✗   | 0.957          | 0.918        | 0.496        | 30.63                |
| +RLL        | ✓                 | ✗   | ✓   | 0.957          | 0.911        | 0.496        | 30.59                |
| +RML        | ✓                 | ✓   | ✗   | 0.970          | 0.919        | 0.491        | 29.57                |
| +RBL        | ✓                 | ✓   | ✓   | <b>0.958</b>   | <b>0.929</b> | <b>0.494</b> | <b>29.37</b>         |

W, achieving a detection accuracy of 0.929 for the second branching point, surpassing both time requirements and the capabilities of previous methods.

Future work should focus on improving detection accuracy while maintaining processing speed through data augmentation and model enhancements. Additionally, implementing the system in actual peeling machines and testing under various conditions is needed to validate its effectiveness.

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